Matthew J. Iacchei

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EDUCATION

2013 University of Hawai'i at Mānoa
Ph.D. Zoology: Ecology, Evolution, and Conservation Biology Specialization
2002 Cornell University, Ithaca, NY
B.S. Biology: Ecology and Evolution, Marine Science; Cum Laude

PROFESSIONAL APPOINTMENTS

2024 – Present	Associate Professor of Marine Science, Hawai'i Pacific University
2025 – Present	Adjunct Graduate Faculty, Biology Department, Texas A&M Corpus Christi
2018 – Present	Affiliate Faculty, Marine Biology Graduate Program, University of Hawai'i Mānoa
2017 – 2023	Assistant Professor of Marine Science, Hawai'i Pacific University
2015 – 2017	NSF Ocean Sciences Postdoctoral Research Fellow, University of Hawai'i Mānoa
2014 – 2015	Lecturer, Department of Math and Sciences, Kapi'olani Community College
2014 – 2015	Postdoctoral Researcher, Biology Department, Texas A&M University, Corpus Christi
2013 – 2014	Postdoctoral Researcher, Oceanography Department, University of Hawai'i Mānoa

PUBLICATIONS (+ denotes a student collaborator)

Submitted

Matur MY+, **Iacchei MJ**. Examining the spatial variability and macroalgal affinities of dinoflagellates affiliated with ciguatera poisoning on O'ahu, Hawai'i. *Journal of Undergraduate Research*. In review.

Laliberte SM+, Mead L, Olayon T, Datlof E, Sutton JT, Kawelo AH, Kotubetey K, Burns JHR, **Iacchei MJ**. A rapid, inexpensive, three-primer genotyping method to distinguish native 'ama'ama (*Mugil cephalus*) from invasive kanda (*Osteomugil engeli*) mullet recruits in Hawai'i. *Conservation Genetics Resources*. In review.

Islas A+, **Iacchei MJ**, Hyrenbach KD. Nearshore prey support skipjack tuna (Katsuwonus pelamis) foraging in Hawaiian waters: Insights from stable isotopes and DNA barcoding. *Frontiers in Marine Science*. In revision.

Lawlor CM+, Suchocki, CR, Lewis CJ, Toonen RJ, Hyrenbach KD, **Iacchei MJ**. Quantifying the effects of colony size, morphology, and bleaching on growth rates of the coral *Montipora capitata* using 3D structure-from-motion photogrammetry. *PeerJ*. In revision.

Published

26. Deck AR, Tucker SJ, Hauser BM, Iacchei MJ, Callan CK, Rappé MS, Nigro OD. 2026. Succession of the Yellow Tang (*Zebrasoma flavescens*) Microbiome from Hatch to Settlement in a Recirculating Aquaculture System. *PeerJ*. In press.

25. Shaw KR, Bramble ZD[+], Royer SJ, Egger M**, Iacchei M**, Lynch JM, Hyrenbach KD. 2026. Plastic ingestion by North Pacific garbage patch fishes: highest occurrences and quantities in upper trophic pelagics. Environmental Research Communications. 8(5):055062.
24. Toonen RJ, **Iacchei MJ**, Bowen BW. 2025. Marine Biogeography. Encyclopedia of Evolutionary Biology. Vol. 2, 18 pp.
23. Freel EB, Conklin EE, Knapp ISS, Kraft DW, Johnston EC, Forsman ZH, Coleman RR, Whitney JL, **Iacchei MJ**, Bowen BW, Toonen, RJ. 2025. Population genetics to population genomics: Revisiting multispecies connectivity of the Hawaiian Archipelago. *Fishes*. 10(12):623. https://doi.org/10.3390/fishes10120623
22. Caldwell IR, Hobbs JP, Bowen BW, Cowman PF, DiBattista JD, Whitney JL, Ahti PA, Belderok R, Canfield S, Coleman RR, **Iacchei M**, Johnston EC, Knapp I, Nalley EM, Staeudle TM, Láruson ÁJ. (2024). Global trends and biases in biodiversity conservation research. *Cell Reports Sustainability*, 1(5), 100082.
21. Nimz I[+], Renshaw MA, Baczenas J[+], Vanderlip C, Hyrenbach KD, **Iacchei M**. (2022). MetaBARFcoding: DNA-barcoding of regurgitated prey yields insights into Christmas shearwater (*Puffinus nativitatis*) foraging ecology at Hōlanikū (Kure Atoll), Hawai'i. *Environmental DNA*, 4(1), 254-268.
20. Nimz I[+], **Iacchei M**, Renshaw MA, Vanderwerf E, Vanderlip C, Saunter M, Worcester N, Hyrenbach KD. (2021). Diving in! First report of the underwater behavior of Christmas shearwaters (*Puffinus nativitatis*) from Kure Atoll. *Marine Ornithology,* 49, 25-35.
19. Riginos C, Bird CE, Liggins L, Treml E, Beger M, Barber PH, Connolly SR, Cowman PF, DiBattista J, Eble JA, Magnuson SF, Horne JB, Kochzius M, Lessios HA, Liu SYV, Ludt WB, Madduppa H, Pandolfi JM, Toonen RJ, **Contributing Members of the Diversity of the Indo-Pacific Network**, Gaither MR (2019). The molecular biogeography of the Indo-Pacific: Testing hypotheses with multispecies genetic patterns. *Global Ecology and Biogeography*. doi:10.1111/geb.12905.
18. Crandall ED, Toonen RJ, **ToBo Laboratory**, & Selkoe KA (2019). A coalescent sampler successfully detects biologically meaningful population structure overlooked by F-statistics. *Evolutionary Applications*, 12(2), 255-265.
17. Dudoit, A.[+], **M. Iacchei,** R.R. Coleman, M.R. Gaither, W.E. Browne, B.W. Bowen, R.J. Toonen. 2018. The little shrimp that could: phylogeography of the circumtropical shrimp, *Stenopus hispidus* (Crustacea: Decapoda), reveals divergent Atlantic and Pacific lineages. *PeerJ* 6:e4409. doi: 10.7717/peerj.4409.
16. **Iacchei**, **M.**, E. Butcher[+], E. Portner, E. Goetze. 2017. It's about time: using a generalized entropy framework to gain insight into temporal genetic patterns in oceanic zooplankton. *Limnology & Oceanography*. doi: 10.1002/lno.10538
15. Selkoe, K.A., C.C. D'Aloia, E.D. Crandall, **M. Iacchei**, L. Liggins, J.B. Puritz, S. von der Heyden, R.J. Toonen. 2016. A decade of seascape genetics: contributions to basic and applied marine connectivity. *Marine Ecology Progress Series*. doi: 10.3354/meps11792.
14. Bowen, B.W., M.R. Gaither, J.D. DiBattista, **M. Iacchei**, K. Andrews, W.S. Grant, R.J. Toonen, J.C. Briggs. 2016. Comparative phylogeography of the ocean planet. *Proceedings of the National Academy of Sciences*. doi: 10.1073/pnas.1602404113
13. Goetze E., P.T. Hudepohl, C. Chang, L. Van Woudenberg, **M. Iacchei**, K.T.C.A. Peijnenburg. 2016. Ecological dispersal barrier across the equatorial Atlantic in a migratory planktonic copepod. *Progress in Oceanography*. doi: 10.1016/j.pocean.2016.07.001
12. **Iacchei M.**, M.R. Gaither, B.W. Bowen, R.J. Toonen. 2016. Testing dispersal limits in the sea: range-wide phylogeography of the pronghorn spiny lobster *Panulirus penicillatus*. *Journal of Biogeography* 43(5): 1032–1044.
11. Selkoe, K.A., O.E. Gaggiotti, E.A. Treml, J.L.K. Wren, M.K. Donovan, **Hawai'i Reef Connectivity Consortium**, R.J. Toonen. 2016. The DNA of coral reef biodiversity: predicting and protecting genetic diversity of reef assemblages. *Proceedings of the Royal Society B* 283(1829): 20160354.

10. Toonen R.J., B.W. Bowen, **M. Iacchei**, J. Briggs. 2015. Biogeography, Marine. In: Kliman, R.M. (ed.), *Encyclopedia of Evolutionary Biology*. vol. 1, pp. 166–178. Oxford: Academic Press.
09. **Iacchei, M.** 2014. Population genetic and phylogeographic insights into the phyllosomal odyssey. *Frontiers in Biogeography* 6(1): 29-37.
08. Selkoe, K.A., O.E. Gaggiotti, **ToBo Laboratory**, B.W. Bowen, R.J. Toonen. 2014. Emergent patterns of population genetic structure for a coral reef community. *Molecular Ecology* 23: 3064-3079.
07. **Iacchei, M.**, J.M. O'Malley, R.J. Toonen. 2014. After the gold rush: population structure of spiny lobsters in Hawai'i following a fishery closure and the implications for contemporary spatial management. *Bulletin of Marine Science* 90(1): 331-357.
06. **Iacchei, M.**, R.J. Toonen. 2013. Caverns, compressed air, and crustacean connectivity: insights into Hawaiian spiny lobster populations. In: Lang, M.A., and M.D.J. Sayer, eds. *Proceedings of the Curacao AAUS/ESDP Joint International Scientific Diving Symposium*. Dauphin Island: American Academy of Underwater Sciences. pp. 115-124.
05. **Iacchei, M.**, T. Ben-Horin, K.A. Selkoe, C.E. Bird, F.J. García-Rodríguez, R.J. Toonen. 2013. Combined analyses of kinship and F_{ST} suggest potential drivers of chaotic genetic patchiness in high gene flow populations. *Molecular Ecology* 22: 3476-3494.
04. Toonen, R.J., K.R. Andrews, I.B. Baums, C.E. Bird, G.T. Concepcion, T.S. Daly-Engel, J.A. Eble, A. Faucci, M.R. Gaither, **M. Iacchei**, J.B. Puritz, J.K. Schultz, D.J. Skillings, M.A. Timmers, B.W. Bowen. 2011. Defining boundaries for ecosystem-based management: a multispecies case study of marine connectivity across the Hawaiian Archipelago. *Journal of Marine Biology*, vol. 2011, Article ID 460173, 13 pages. doi: 10.1155/2011/460173
03. Selkoe, K.A., J.R. Watson, C. White, T. Ben-Horin, **M. Iacchei**, S. Mitarai, D.A. Siegel, S.D. Gaines, R.J. Toonen. 2010. Taking the chaos out of genetic patchiness: seascape genetics reveals ecological and oceanographic drivers of genetic patterns in three temperate reef species. *Molecular Ecology* 19: 3708-3726.
02. Ben-Horin, T., **M. Iacchei**, K.A. Selkoe, T.T. Mai, R.J. Toonen. 2009. Characterization of eight polymorphic microsatellite loci for the California spiny lobster, *Panulirus interruptus* and cross-amplification in other achelate lobsters. *Conservation Genetics Resources* 1: 193-197.
01. **Iacchei, M.**, P. Robinson, K.A. Miller. 2005. Direct impacts of commercial and recreational fishing on spiny lobster populations at Santa Catalina Island, California, United States. *New Zealand Journal of Marine and Freshwater Research* 39: 1201–1214.

Technical Reports and Other Publications

Iacchei, M., K.M. Poepoe. 2015. Population genetic structure of the pronghorn spiny lobster *Panulirus penicillatus* at Mo'omomi, Moloka'i, HI. A Report for the Western Pacific Fisheries Management Council. 21p.

Iacchei, M. 2012. The impact of 'opihi on intertidal communities. Ka Pili Kai, University of Hawai'i Sea Grant College Program Publication 34 (1): 3-5.

Iacchei, M. 2011. Managing fisheries for the future. Ka Pili Kai, University of Hawai'i Sea Grant College Program Publication 33 (1): 3-5.

GRANTS AND FELLOWSHIPS

Awarded, but not commenced

Jan 2027 – 2029 **Inflation Reduction Act: NOAA Fisheries Citizen Science Funding ($59,890 to HPU)**
Understanding spawning of pelagic species through fisher plankton collection

PI: Cassandra Pardee at Poseidon Fisheries Research; $137,786 total, with ~$59,890 subcontracted to my lab for student support to complete the genetic identification of fisher-collected plankton.

Jan 2027 – 2030 **NOAA Coral Reef Conservation Program (~$40,000 to HPU)**
Lobster surveys to assess Maui's lobster fishery
PI: Cassandra Pardee at Poseidon Fisheries Research; $240,723 total, with ~$40,000 for student summer support and inter-island travel, and all supplies covered in addition.

Jan 2027 – 2029 **NOAA Saltonstall-Kennedy Research Grant (~$140,139 to HPU)**
Species composition, biomass, and seasonality at fish aggregating devices in Maui, HI
PI: Cassandra Pardee at Poseidon Fisheries Research; $391,205 total, with ~$140,139 subcontracted to my lab for student support and eDNA extraction, library preparation and sequence analysis

Completed

2024 – 2025 **National Institute of Health INBRE Pilot Project ($64,189)**
Building capacity to estimate the spatiotemporal variability and environmental drivers of *Gambierdiscus* spp. composition to improve food security in Hawai'i

2024 – 2025 **Hawai'i Division of Aquatic Resources Contract ($147, 992)**
Preliminary assessment of genetic diversity of wild stock 'ama 'ama mullet (*Mugil cephalus*) in Hawai'i

2022 – 2023 **HPU College of Natural Sciences Scholarly Endeavors Program Grant ($10,000)**
DNA barcoding and dDNA metabarcoding of aku and ahi tuna gut contents

2022 – 2023 **Oikonos-Ecosystem Knowledge Collaborative Grant ($13,000)**
DNA barcoding and dDNA metabarcoding of aku and ahi tuna gut contents

2019 – 2021 **Hawai'i Fish Habitat Partnership ($29,440)**
Environmental DNA detection of ecological community change after mangrove removal from Honouliuli stream mouth

2019 – 2020 **Hawai'i Division of Aquatic Resources Contract ($60,000)**
East Maui Watershed stream and estuarine community monitoring using environmental DNA metabarcoding; Co-PI with Mark Renshaw, Oceanic Institute

2018 – 2019 **HPU College of Natural Sciences Scholarly Endeavors Program Grant ($5,000)**

2017 – 2018 **HPU College of Natural Sciences Scholarly Endeavors Program Grant ($2,500)**

2015 – 2017 **NSF Ocean Sciences Postdoctoral Research Fellowship ($175,000)**
What drives population connectivity in the open sea?

2015 – 2018 **NOAA Saltonstall–Kennedy Grant ($399,516; Declined)**
Testing the Effects of Community-managed "Rest Areas" on Coastal Hawaiian Fisheries
Co-authored with PI C.E. Bird (TAMUCC); Declined post-doc, had started NSF PRF

2014 – 2015 **Pacific Gas and Electric Grant ($45,000)**
Population genomics of white seabass (*Atractoscion nobilis*)
Co-PI with H.S. Lenihan (UCSB), K.A. Selkoe (NCEAS), and R.J. Toonen (UH); Managed UH staff and portion of the funds ($41,750)

2014 **Student Activities Program Board Grant ($1,869)**
Funding for the Albert L. Tester Symposium; Co-PI with N.J. Silbiger and J. Whitney

2013 – 2014 **Western Pacific Regional Fishery Management Council Grant ($25,000)**
Connectivity of the lobster (*Panulirus penicillatus*) population at Mo'omomi, Moloka'i to the other Hawaiian Islands
Co-PI with Kelson "Mac" Poepoe and Hui Malama O Mo'omomi

2012 **Jessie D. Kay Research Grant ($10,000)**
Effects of 'opihi density and size structure on intertidal community diversity

2012 **Student Activities Program Board Grant ($3,724)**
Funding to host a Software Carpentry Workshop; Co-PI with M.C. Siple

2011 **Travel Grant ($746)**
Graduate Student Organization, University of Hawai'i

2009 – 2012 **Sea Grant Fellowship (~$191,000; - salary, tuition, $20,000 yearly research budget)**
An empirical test of ecosystem-based management: how density and size structure of 'opihi affect community structure
Co-authored with PI R.J. Toonen

2009 **Charles H. and Margaret B. Edmondson Research Grant ($1,500)**
Population connectivity of spiny lobsters throughout the Hawaiian archipelago

2008 **Travel Grant ($600)**
Ecology, Evolution and Conservation Biology Graduate Group, University of Hawai'i

2007 **Travel Grant ($675)**
Ecology, Evolution and Conservation Biology Graduate Group, University of Hawai'i

2007 **Watson T. Yoshimoto Grant ($2,500)**
Population connectivity of spiny lobsters throughout the Hawaiian archipelago

2006 – 2009 **EPA STAR Fellowship (~$111,000 – salary, tuition, $5,000 yearly research budget)**
Management without borders: population connectivity of the "California" spiny lobster (*Panulirus interruptus*) in California, USA and Baja California, Mexico

2006 **Charles H. and Margaret B. Edmondson Research Grant ($800)**
Population connectivity of spiny lobsters throughout the Hawaiian archipelago

2001 **NSF Research Experience for Undergraduates (REU)**
Tropical Marine Science, Cornell University

AWARDS

2022 Kavli Fellow of the National Academy of Sciences; Indonesian-American Symposium
2021 Marine Science Faculty Member of the Year (from HPU student body)
2013 Best Dissertation Award, International Biogeography Society ($500)
2013 Best Presentation, 38th Albert L. Tester Memorial Symposium ($1,000)
2010 Best Presentation, Honorable Mention 35th Albert L. Tester Memorial Symposium
2006 NSF Graduate Research Fellowship Program (GRFP) – Honorable Mention

TEACHING

Instructor of Record – Hawai'i Pacific University

Marine Ecology (Fall, 2022 - Present)
4000 and/or 6000-level lecture course for senior marine science majors and MSMS graduate students; 25-50 students. I have taught this as two separate courses and as a combined course.

Marine Biology and Biological Oceanography Lecture and Lab (Spring, 2018 – Present)
3000-level lecture course for majors with ~40-60 students; 3-4 labs with 10-12 students each).

Bioinformatics (Fall 2019 – Present; odd years)
6000-level course for Marine Science Master's students focused on analysis of genetic and genomic data. Co-taught with Olivia Nigro, Ph.D; 9 -12 students.

Marine Invertebrates Laboratory (Spring 2024, 2025)
3000-level lab course for marine science majors (10 – 12 students)

Sustainability in Galápagos and Hawai'i (Summer 2023, 2024)

Interdisciplinary field course for undergraduate and graduate students, co-taught with HPU Anthropology Professor Adam Burke; 10 - 12 students, 3-week summer course in Galápagos

Guest Speakers in Marine Science Seminar (Fall 2026, 2024)
6000-level course for MSMS graduate students showcasing Hawai'i researchers and their work

Professional Development and Communicating Ocean Science (Spring, 2018 – 2023)
6000-level course for Applied Marine Science Master's Program students; 8 - 12 students.

Marine Science Senior Capstone (Spring 2020)
4000-level course in which graduating students complete an independent research project, write it up in a journal format, and present their results at a department symposium; 8 students this year

Oceanographic Field Techniques Lab (Fall 2018)
1000-level field course focused on preparing first and second-year students to effectively conduct research on a small research vessel (42-foot motorized catamaran); 3 sections of 12 students each.

Chemical and Geological Oceanography Lab (Fall 2017, 2019)
3000-level field and laboratory course focused on geological and chemical oceanography; four sections of 10 - 12 students each.

Instructor of Record –University of Hawai'i System

Quantitative Underwater Ecological Survey Techniques, University of Hawai'i (Summer 2019, 2026)
Co-instructor with a large team led by Lisa Parr, Jeff Kuwabara, and Matt Connelley. Two-week summer field course in Ke'ei, Hawai'i Island for AAUS divers and divers in training. I was a team lead and taught the invertebrates and statistics lectures.

Introductory Biology I, Kapi'olani Community College (Spring 2014)
First semester introductory biology course for ~30 students. Course content included cellular and molecular biology, genetics, evolution, and the history of life.

UH@SEA: A Field Course in Marine Conservation, University of Hawai'i at Mānoa (Summer 2010)
Co-Instructor with Prof. Alison Rieser, Dr. Cynthia Hunter, Dr. David Hyrenbach, and Dr. Jan Witting. Five-week marine conservation course, with three weeks conducting field research throughout Hawai'i while onboard the Sea Education Association tall ship Robert C. Seamans. Developed multi-disciplinary course lessons, led intertidal and subtidal field excursions, supervised student research projects, and conducted student evaluations.

Teaching Assistant

Spring 2012 **Advanced Biometry**; University of Hawai'i at Mānoa; Course Professor: Dr. Andy Taylor
2000 – 2001 **Biochemistry**; Cornell University; Course Professor: Jim Blankenship

Invited Seminars and Guest Lectures

2024 Kids characterizing coastal communities: A collaborative approach to monitoring nearshore fish diversity on Hawai'i Island using eDNA
Biology Department Seminar, Universidad de Vigo, Spain

2022 Kids characterizing coastal communities: A collaborative approach to monitoring nearshore fish diversity on Hawai'i Island using eDNA
Marine Science Seminar Series, Hawai'i Pacific University

2018 What is the scale of population connectivity in marine ecosystems?
Universidad Autónoma de Baja California Sur

2018 It's about time: Insights into temporal genetic patterns from biodiversity indices.
Marine Biology Seminar Series, University of Hawai'i at Mānoa

2017 Comparative biogeography and phylogeography of Indo-Pacific marine invertebrates
Universidad Autónoma de Baja California Sur

2017 Genetic connectivity of marine invertebrate populations throughout Hawai'i

Zoology 450: Natural History of the Hawaiian Islands. University of Hawai'i at Mānoa

2016 How to conduct research in Hawai'i
MBio 603: Career Skills for Marine Biologists. University of Hawai'i at Mānoa
Co-taught with Kaleonani Hurley and Kanoe Morishige

2016 Population connectivity of marine invertebrates in Hawai'i
Zoology 450: Natural History of the Hawaiian Islands. University of Hawai'i at Mānoa

2015 Testing dispersal limits in the sea: range-wide phylogeography of *Panulirus penicillatus*
Hawai'i Institute of Marine Biology Seminar Series

2015 A tail of two spiny lobsters: connectivity of *Panulirus marginatus* and *P. penicillatus* in Hawai'i
Introductory Marine Biology. Kapi'olani Community College

2014 Defining Boundaries for Ecosystem-Based Management
Biol 602: Marine Biology – Processes and Impacts. University of Hawai'i at Mānoa

2013 Patterns of Genetic Connectivity in Marine Invertebrates in Hawai'i
Introductory Marine Biology. Kapi'olani Community College

2012 Self-recruitment in the California spiny lobster (*Panulirus interruptus*)
NMFS Southwest Fisheries Science Center

2010 Deep seas and hydrothermal vents
Biology 301: Marine Ecology and Evolution. University of Hawai'i at Mānoa

2010 The classic model of ocean ecosystems: of phytoplankton and copepods
Biology 301: Marine Ecology and Evolution. University of Hawai'i at Mānoa

2010 New model of ocean ecosystems: food webs at the nanoscale
Biology 301: Marine Ecology and Evolution. University of Hawai'i at Mānoa

2004 California spiny lobster: ecology, population biology, and fisheries at Santa Catalina Island
Achievement Rewards for College Students (ARCS) Foundation, Inc.

2004 Biology, ecology, and fisheries management of the California spiny lobster
East/West Marine Biology Program, Wrigley Marine Science Center

2004 Research methodology for kelp forest fisheries ecology
East/West Marine Biology Program, Wrigley Marine Science Center

MENTORING

Master's Students, Hawai'i Pacific University – Committee Chair/Co-Chair (N=10) (graduation year)
Liana Gaspar (current); Cortney Deterding (applied track; current); Irie Baniqued (current); Jonathan Lin (current); Sophia Samus (current); Alondra Reyes (2026); Klara Sobotikova (2025); Sydney Luitgaarden (applied track, co-supervised with D. Hyrenbach; 2024); Colleen Lawlor (2024); Savannah Laliberte (dng); Nan Himmelsbach (2021)

Master's Students, Hawai'i Pacific University – Committee Member (N=13)
Ava Dufort (current); Emmylou Kidder (current); Jonah Sheridan (current); Julia Hunckler (dnf); Elena Polizotto (2025); Alyssa Piauwasdy (2024); Annie Deck (2024); Hanna Mantanona (2023); Kyla Richards (2023); Alondra Islas Vazquez (2022); Cherise Spotkaeff (2022); Anita Harrington (2020); Illana Nimz (2020)

Doctoral Students, University of Hawai'i – Committee Member (N=2)
Spencer Davis, SOEST (current); Nakoa Goo, SOEST (current)

Master's Students, Texas A&M Corpus Christi – Committee Member (N=1)
Keenan Larivierre (current)

Undergraduate Students, Hawai'i Pacific University – Senior Capstone Project Mentor or funded student (N=21)

Emily Bandalos (current); Kaylee Tanig (current); Leila Tewahade (2026); Hikari Bauson (2025); Amelia Hart (2025); Aspen Pawlowski (2025); Veronika Pearson (2025); Diya Reddy (2025); Valentina Salgado-Mondanez (2025); Caitlyn Visser (2025-transferred); Mylou Matur (2024); Alondra Reyes (2024); Roberta Cardenas (2023); Madison Gramse (2022); Victor Lopez (2022); Daniel Mendoza-Hermosillo (2022); Sophia Samus (2021); Sara Stiverson (2021); Catalina Ferrari (2020); Savannah Laliberte (2019); John Baczenas (2018)

Post-doctoral Project Supervisor and Mentor for Undergraduate and Graduate Students at University of Hawai'i at Mānoa

Jonathan Whitney (Ph.D. student, 2015-17); William Hoaka Thomas (undergraduate, 2015-17); Stephanie Matthews (undergraduate, 2014-15); Michelle Uchida (undergraduate, 2014); Lauren Van Woudenberg (M.S. Student, 2013-16); Kaleonani Hurley (Ph.D. student, 2013-17); Elizabeth Butcher (undergraduate, 2013-14); Alexander Shiarella (Le Jardin High School senior, 2011)

Additional Mentoring

Lead Faculty Advisor for undergraduate marine science majors at HPU – advise 45 – 65 students annually (primarily junior and seniors) on their academic pathway to ensure they graduate in four years, or as early as possible, and assist them with finding post-graduate career opportunities (2018 – Present)

Grant proposal workshop – co-lead with one or two other faculty members each fall when we have more than one new student interested in preparing NSF GRFP or other graduate student fellowship proposals. We have had one HPU student receive a GRFP and three receive honorable mention. (Fall 2018, 2020, 2021, 2022, 2023, 2026)

Writing mentor for three PhD students and one undergraduate to teach them about the publication process by writing and peer-reviewing manuscripts, UH Månoa (2017 - 2018)

Project mentor for upper-level experiential marine biology class project: Differential response of two 'opihi species to various intertidal predators, Hawai'i Pacific University (2011)

Project mentor for student group project on ecological and human dimensions of the Hawaiian 'opihi (*Cellana* sp.) fisheries, UH@SEA, UH Mānoa (2010)

COMMUNITY ENGAGEMENT

2025	HPU on Tap Public Presentation at Spaghetti Factory: "Ciguatera Non Grata! Figuring out the biology of small dinoflagellates to help prevent fish poisoning in Hawai'i"
2024	Presentation and informational flyer on ciguatera poisoning for Waimānalo Pono Research
2024 – Present	Waimānalo Pono Research Member
2024 – 2025	Ke Kula Nui O Waimānalo Lawai'a Camp: transect and quadrat survey training and strawberry DNA lab instruction
2023	Assistend with hosting and demonstrating a strawberry DNA lab for students from Nāwahī Ko Kula Kai Program
2021 – 2023	Project Advisory Group Member – He'eia NERR/PICASC eDNA monitoring of King Tide effects on nearshore community composition
2022	Selected interviewee (with the HPU Marine Science Education Coordinator, Gary Karr) for a Stanford University dissertation project: "eDNA for Community Science"
2019, 2020	Wai'anae High School Class lecture and lab demonstration: "Using eDNA to detect coral reef community change through time"

2019	Kua O Ka Lā New Century Public Charter School: eDNA for community fish monitoring Trained teachers to use a Mini-PCR machine purchased for eDNA class projects
2018	Public talk and participatory science program – Rocio del Mar liveaboard dive boat
2017	ʻOpihi Summit Workshop Participant and Trainer for intertidal survey methods
2016	Lab tour and zooplankton lessons – Wilson Elementary School 6th grade class
2015	Ula (lobster) fact sheet for Alahula ʻĀina Momona (Pathways to Abundance) efforts to support community based marine stewardship
2013	Dissertation press coverage: Phys.org http://phys.org/news/2013-10-family-doesnt-sea-creature.html
2013	Volunteer – zooplankton lab at the School of Ocean and Earth Science and Technology (SOEST) open house at the University of Hawaiʻi at Mānoa
2011	Class lecture and lab demonstration – Roosevelt High School AP Biology
2011	Public talk – Hanauma Bay Education Program
2011	Science judge – Ocean Science Bowl, University of Hawaiʻi at Mānoa
2010	Class Lecture and Lab – Nalu Studies Program
2010	Crustacean expert – Family Night at the Waikiki Aquarium
2010	Public talk – Hanauma Bay Education Program
2009 – Present	ʻOpihi Partnership collaborator – survey intertidal habitats to monitor ʻopihi populations with local communities, school groups, NGOs, and state and federal agencies.

PROFESSIONAL SERVICE

Review Editor: Frontiers in Marine Science

Journal Reviews: Atoll Research Bulletin, Bioinvasions, Bulletin of the Southern California Academy of Sciences, Continental Shelf Research, Coral Reefs, Ecology and Evolution, Fish and Fisheries, Heredity, Hydrobiologia, ICES Journal of Marine Science, Journal of Biogeography, Journal of Crustacean Biology, Journal of Marine Biology, Marine Ecology Progress Series, Molecular Ecology, Molecular Ecology Resources, New Zealand Journal of Marine and Freshwater Research, Pacific Science, PeerJ, PLoS ONE, PNAS

Grant Reviews: California Sea Grant, NMFS MARFIN Program, National Science Foundation (NSF) Bio-OCE (*ad hoc* reviewer)

Activities

2025	External Thesis Reviewer – Duke Paane, Global Environmental Science, UH Mānoa
2025	Pūlama Kuamoʻo Mentorship Program, Hawaiʻi Conservation Conference
2023	External Tenure Reviewer – Biology Faculty Member, BYU Hawaiʻi
2023 – Present	HPU IACUUC Committee member
2020 – Present	HPU Dive Control Board member
2020	Student presentation judge, Hawaiʻi Conservation Conference
2019 – 2020	National Marine Educator Conference Committee Member
2018, 2019	Invited Panelist, Alpha Chi Honor Society "Preparing For Graduate School" Symposium
2018	International external reviewer, University of Tasmania Ph.D. dissertation
2017	Student presentation judge, Larval Biology Conference
2017	Meeting mentor for two graduate students – ASLO Meeting
2016	Student presentation judge, Seascape genetics session, International Coral Reef Symposium
2013 – 2014	Co-organizer, 39th Albert L. Tester Memorial Symposium, University of Hawaiʻi at Mānoa

2012 Co-organizer, Software Carpentry 2-day workshop, University of Hawai'i at Mānoa
2012 Invited participant, SeaHarmony Focus Group
2012 Invited panelist, "Mentoring, a key to success in graduate school" organized by the Center for Teaching Excellence, University of Hawai'i at Mānoa
2011 Student committee, The Crustacean Society Meeting, Honolulu, HI
2006 Graduate student representative, Department of Zoology, University of Hawai'i at Mānoa
2006 Zoology Department representative to the Graduate Student Organization (GSO)
2004 External review board – California spiny lobster group masters thesis, Donald Bren School of Environmental Science & Management, University of California, Santa Barbara
2001 – 2002 Academic advisor for undergraduates, Department of Biology, Cornell University
2001 – 2002 Student member of Academic Advising Committee, Shoals Marine Laboratory

CONFERENCE PRESENTATIONS (* denotes poster)

2026 International Coral Reef Symposium. *An integrative approach to understanding risk landscapes for ciguatera poisoning in Hawai'i**
2026 International Coral Reef Symposium. *Quantifying the Effects of Colony Size, Morphology, & Bleaching on Growth Rates of the Coral* Montipora capitata *Using 3D Structure-from-Motion Photogrammetry*
2026 International Coral Reef Symposium. *Spatiotemporal Dynamics and Larval Dispersal of Spiny and Slipper (Achelata) Lobster Phyllosoma Larvae in the Hawaiian Archipelago**
2026 International Coral Reef Symposium. *Spatio-temporal variability of ichthyoplankton in Kāne'ohe Bay, O'ahu**
2025 Pacific Islands Chapter of the American Fisheries Society – Inaugural Meeting. *Assessing the genetic diversity of wild 'ama'ama (Mugil cephalus) populations in Hawai'i to inform restocking efforts*
2022 Managing Biodiversity and Conservation for Sustainable Development in Tropical Ecosystems International Conference (MANBIOCONTE2). **Invited Plenary Speaker.** *Kids characterizing coastal communities: A collaborative approach to monitoring nearshore fish diversity on Hawai'i Island using eDNA*
2022 Indonesian-American Kavli Frontiers of Science Symposium. **Invited Participant.** Presentation: *Kids characterizing coastal communities: A collaborative approach to monitoring nearshore fish diversity on Hawai'i Island using environmental DNA*
2020 Hawai'i Conservation Conference. **Invited Talk and Panelist**. *Innovations in fish diversity assessment methodology in Hawai'i streams and estuaries*
2020 Association for the Sciences of Limnology and Oceanography Meeting. **Invited Talk.** *Using Ocean Observing Initiative (OOI) data to enhance student learning about factors affecting primary production*
2018 Australian Marine Science Association (AMSA) Conference. **Invited Talk.** *Genomically-informed fisheries management of white seabass (*Atractoscion nobilis*)*
2018 5th International Marine Conservation Congress. *Genomically-informed fisheries management of white seabass (*Atractoscion nobilis*)*
2017 Larval Biology Meeting. **Invited Plenary Talk.** *Kinsights into chaotic genetic patchiness in high gene-flow populations*
2017 Evolution Meeting. *It's about time: Insights into temporal genetic patterns from biodiversity indices.*
2017 Association for the Sciences of Limnology and Oceanography Meeting. *Exploring the environmental drivers of genomic differentiation in a mesopelagic copepod,* Pleuromamma xiphias.
2016 International Coral Reef Symposium. *Testing dispersal limits in the sea.*
2016 American Society of Naturalists. *Identifying environmental drivers of genetic differentiation in the open sea.**
2014 Hawai'i Conservation Conference. *Population structure of spiny lobsters in Hawai'i following a fishery closure, and the implications for contemporary spatial management.*
2014 Asian Pacific Coral Reef Symposium. *Testing provincial boundaries for phyllosoma: Range-wide phylogeography*

of the pronghorn spiny lobster Panulirus penicillatus.

2014 Ocean Sciences Meeting, Association for the Sciences of Limnology and Oceanography. *Kinship metrics provide insight into the murky population genetic datasets of high gene-flow populations.*

2013 Albert L. Tester Memorial Symposium. *Combined analyses of kinship and F_{ST} suggest potential drivers of chaotic genetic patchiness in high gene-flow populations.* **Best Paper.**

2012 Western Society of Naturalists. *Self-recruitment in the California spiny lobster (*Panulirus interruptus*) despite an extremely long pelagic duration.*

2012 Albert L. Tester Memorial Symposium. *Can fisheries regulations drive ecological community change?*

2011 American Fisheries Society. *Can harvesting regulations drive ecological community change?*

2011 American Fisheries Society. *Management without borders: population genetics of the "California" spiny lobster*

2011 Ecological Society of America. *Can harvesting regulations drive ecological community change?: Effects of limpet (*Cellana sandwicensis*) size and density on intertidal algal communities.*

2010 Western Society of Naturalists. *Does size matter, or is more always better?: Effects of limpet (*Cellana sandwicensis*) size and density on intertidal communities.**

2010 Annual American Academy of Underwater Sciences Symposium. *Caverns, compressed air, and crustacean connectivity: insights into Hawaiian spiny lobster populations.*

2010 Albert L. Tester Memorial Symposium. *A tail of two spiny lobsters: population connectivity of* Panulirus marginatus *and* Panulirus penicillatus *in Hawai'i.*

2009 International Marine Conservation Congress. *Management without borders: population genetics of the "California" spiny lobster (*Panulirus interruptus*).*

2009 Pacific Science Inter-Congress. *Phylogeography of an Hawaiian endemic spiny lobster,* Panulirus marginatus*: Implications for management and biodiversity conservation.*

2008 Albert L. Tester Memorial Symposium. *Population genetics and international fisheries management: the "California" spiny lobster (*Panulirus interruptus*) story.*

2007 Western Society of Naturalists. *Management Across Borders: An mtDNA perspective of the "California" spiny lobster,* Panulirus interruptus.

2004 Western Society of Naturalists. *The effects of two different marine reserve designs on spiny lobster populations at Santa Catalina Island, CA.*

2004 International Lobster Biology and Management Conference. *Direct impacts of commercial and recreational fishing on spiny lobster populations at Santa Catalina Island, California, United States.*

2003 Western Society of Naturalists. *The impacts of commercial and recreational fishing on spiny lobster populations at Santa Catalina Island, CA.*

2003 Western Society of Naturalists. *The ecology and population structure of adult California spiny lobster at Santa Catalina Island, CA.**

2001 Centro Ecologico Akumal Research Symposium. *Severity and rate of increase of bleaching of* Siderastrea siderea *at three depths in Akumal, Mexico.*

OTHER RELEVANT CERTIFICATIONS AND EXPERIENCE

Boating and Diving

2010 – U.S. Dept. of the Interior and Dept. of Commerce Motorboat Operator Certification
2007 – Small boat operator – Hawai'i Institute of Marine Biology
2001 – AAUS scientific diver, master diver, nitrox-certified (500+ research dives)
2001 – First aid/CPR/Oxygen administration
2003 – 2005 Resident volunteer – Catalina Hyperbaric Chamber
2003 – 2005 Small boat operator – University of Southern California

Fieldwork

In addition to expeditions listed above, I have led and participated in field expeditions ranging from three days to one month to conduct intertidal and subtidal research (up to 150 ft depth) on Kauaʻi (5+ trips), Maui (5+ trips), Molokaʻi (2), Hawaiʻi (20+ trips), Palmyra Atoll, Tahiti and Moʻorea, Papua New Guinea, Islas Revillagigedos and Baja California Sur, Mexico, and an oceanographic research cruise on board the Schmidt Ocean Institute's R/V Falkor.

Specialized Training

2016 – Genetic and genomic data analysis in R course, PR Statistics
2016 – Hierarchical Bayesian modeling in R course, PR Statistics
2015 – Software Carpentry workshop: Unix shell scripting, Git version control, Python
2011 – Consortium for Ocean Science Exploration and Engagement (COSEE) Communicating Ocean Sciences workshop

Professional Organizations

American Academy of Underwater Sciences
International Coral Reef Society
National Marine Educators Association
Sigma Xi